

$$5. k = \frac{3}{2}, j = -\frac{6}{1}$$

$$4. \mathbf{r} = (1, 6, 0) + t(1, 1, 0)$$

$$3. (a) 7x + y - 7 = 0$$

$$(b) 2x - 3y - 2 = 0$$

$$(d) \mathbf{r} = (4, 7, -3) + t(1, 0, 0)$$

$$(c) \mathbf{r} = (1, 4, 1) + t(1, 4, 1)$$

$$(b) \mathbf{r} = (0, 1) + t(7, 2)$$

$$2. (a) \mathbf{r} = (3, 4) + t(4, -1)$$

$$\frac{-3}{2} (=t)$$

$$z = 3 - 3t, \frac{x-6}{2} = \frac{y-1}{-1}$$

$$x = 2 + 6t, y = 1 + 2t$$

$$(i) \mathbf{r} = (2, 1, 3) + t(6, 2, -3)$$

$$\frac{4}{3} (=t)$$

$$z = -3 + 4t, \frac{x-1}{4} = \frac{y-4}{-1}$$

$$x = 1 + 4t, y = 4 + 2t$$

$$(h) \mathbf{r} = (1, 4, -3) + t(4, 2, 4)$$

$$\frac{1}{3} = \frac{x-1}{-3} = \frac{y-4}{2} (=t)$$

$$z = 5 + 6t$$

$$x = 1 + t, y = 3 + 3t$$

$$(g) \mathbf{r} = (-4, 3, 5) + t(1, 3, 6)$$

$$x = -4, y = 2, z = 1 + t$$

$$(f) \mathbf{r} = (-4, 2, 1) + t(0, 0, 1)$$

$$\frac{-6}{2} (=t)$$

$$x = 1 + 2t, y = 2 - 6t, z = t$$

$$(e) \mathbf{r} = (1, 2, 0) + t(2, -6, 1)$$

$$y = 2 + t, \frac{x+3}{4} = \frac{z-1}{-2} (=t)$$

$$x = -3 + 4t$$

$$(d) \mathbf{r} = (-3, 2) + t(4, 1)$$

$$y = 1 + 3t, \frac{x-2}{-1} = \frac{z-1}{3} (=t)$$

$$\mathbf{r} = (2, 1) + t(1, 3)$$

$$y = 1 + 4t, \frac{x-6}{-6} = \frac{z-1}{4} (=t)$$

$$x = 6 + 11t$$

$$(c) \mathbf{r} = (6, 1) + t(1, 4)$$

$$y = 3 - t, \frac{x-1}{3} = \frac{z-1}{-3} (=t)$$

$$x = 1 + 3t$$

$$(a) \mathbf{r} = (1, 3) + t(3, -1)$$

Exercise 7.12.

Review Exercise 7.12

- B 1. Find vector, parametric, and (where possible) symmetric equations for the following lines.

- the line through the points (1, 3) and (4, 2)
- the line through the points (6, 1) and (-5, -3)
- the line through the point (2, 1) with direction vector (1, 3)
- the line through the point (-3, 2) and parallel to the line through the points (1, 1) and (5, 2)
- the line through the points (1, 2, 0) and (3, -4, 1)
- the line through the point (-4, 2, 1) and parallel to the z axis
- the line through the point (1, 3, 5) with direction vector (1, 3, 6)
- the line through the point (1, 4, -3) and parallel to the line through the points (6, 1, 1) and (2, -1, -3)
- the line through the point (2, 1, 3) with direction numbers (6, 2, -3).

2. Write a vector equation for the line through the given point parallel to the given line.

- $P(3, 4)$ and $\mathbf{r} = (5, 6) + t(4, -1)$
- $P(0, 1)$ and the line through $A(4, 3)$ and $B(-3, 1)$
- $P(1, 4, 1)$ and the line through $A(-1, 1, 2)$ and $B(0, 5, 3)$
- $P(4, 7, -3)$ and the x axis

3. Write a scalar equation for the line through the given point perpendicular to the given line.

- (1, 0) and $\mathbf{r} = (-4, 3) + t(7, 1)$
- (4, 2) and the line $3x + 2y - 4 = 0$

4. If a line L is defined by $\mathbf{r} = (1, 6, 0) + s(1, -1, 2)$, write a vector equation for the line in the xy plane which is perpendicular to L and passes through (1, 6, 0).

5. $L_1: \mathbf{r} = (2, 1, 3) + t(6, -4, 1)$

$$L_2: \frac{x-1}{-1} = \frac{y-6}{k} = \frac{z-2}{j}$$

For what values of k and j will L_1 be parallel to L_2 ?

6. Find a scalar equation for each of the following planes.

- the plane through the point (1, 2, -6) and with normal $\mathbf{n} = (1, 3, -2)$
- the plane through the point (4, -2, 1) and perpendicular to the line joining the points (2, -1, 2) and (3, 2, -1)
- the plane through the points (2, 0, 1) and (-1, 2, 0) and perpendicular to the plane $2x - 4y - z = 7$
- the plane through the points (2, 3, -1), (-1, 5, 2), and (-4, -2, 2)
- the plane perpendicular to the x axis and containing the point (1, 4, 2)
- the plane through the points (1, 4, 2) and (2, 3, 1) and parallel to the line $\mathbf{r} = 2t\mathbf{i} + (4t + 1)\mathbf{j} + (t - 3)\mathbf{k}$

7. Find vector and parametric equations for each of the planes of Question 6 (c), (d), (f), (g), (h).

8. Find the point of intersection of the line

$$\frac{x+1}{-4} = \frac{y-2}{3} = \frac{z-1}{-2}$$

and the plane $x + 2y - 3z + 10 = 0$.

9. Find both vector and scalar equations for the plane through $A(2, 1, 0)$, $B(-3, 9, 2)$ and $C(5, 2, 1)$.

10. Find the intersection of $L_1: \mathbf{r} = (3, 1, 2) + p(-1, 1, 2)$ and each of the following.

- (a) the line $\mathbf{r} = (2, 3, -1) + b(1, 2, 3) + c(1, 0, 1)$
 (b) the plane containing the origin and the line $L_2: \mathbf{r} = (1, 4, 1) + t(2, -1, 3)$

11. Determine whether $L: \mathbf{r} = (1, 1, 6) + q(2, 6, -4)$ is perpendicular to each of the following planes.

- (a) $\mathbf{r} = (2, 0, 1) + b(1, 1, 2) + c(0, 2, 3)$
 (b) $x + 3y - 2z = 1$

12. (a) Find the equation of the line through the point $(3, 2, -1)$ and perpendicular to the plane $4x + 3y - z + 5 = 0$.
 (b) Find the point of intersection of this line and the given plane.
 (c) Find the distance of the point $(3, 2, -1)$ from the plane.

13. State whether or not each of the following pairs of planes intersect. If the planes do intersect, find the equation of their line of intersection.

- (a) $2x + y - z + 4 = 0$ (c) $3x + y + 4 = 0$
 $4x + 2y - 2z + 7 = 0$ $2x - y + 3z = 1$
 (b) $2x + y - z + 4 = 0$ (d) $2x + y = 2$
 $x + y + 2z + 4 = 0$ $4y + z = 3$

14. In each of the following use normal vectors to discuss the intersection of the planes determined by the equations. State whether the system of equations has a unique solution, no solution, or an infinite number of solutions.

- (a) $x + 2y + 3z = 3$ (c) $4x + 6y - z = 16$
 $2x - 3y + z = 8$ $3x - y = 5$
 $x - y + 4z = 7$ $x + 7y - z = 11$

- (b) $2x + y - z = 3$ (d) $x + y + z = 5$
 $2x + y - z = 8$ $2x - y + 2z = 8$
 $4x + 2y - 2z = 8$ $x - 2y + z = 0$

15. Find the point of intersection of the following pairs of lines.

- (a) $\mathbf{r} = (t + 1)\mathbf{i} + (2t - 3)\mathbf{j}$, with
 $\mathbf{r} = (2s - 1)\mathbf{i} + (-4s - 2)\mathbf{j}$

- (b) $x = t + 2$, $y = 3t + 4$, with
 $x = 2s + 3$, $y = 4s - 1$.

- (c) the line of intersection of $2x - y - z = 3$ and
 $2x + y + 2z = -1$
 with the line of intersection of $3x - y + z = 0$ and
 $2x - 2y - z = 2$

14. (a) unique solution $(2, -1, 1)$; three planes intersect in one point
 (b) no solutions; three planes are parallel
 (c) infinite number of solutions; planes intersect in one line
 (d) no solutions; three planes form a triangle

13. (a) no
 (b) $x = 3t$, $y = -5t - 4$, $z = t$
 (c) $x = -\frac{5}{3}t - \frac{5}{11}$, $y = \frac{5}{3}t - \frac{5}{11}$, $z = t$
 (d) $x = t$, $y = -2t + 2$, $z = 8t - 5$

12. (a) $\mathbf{r} = (3, 2, -1) + t(4, 3, -1)$
 (b) $\left(-\frac{9}{13}, -\frac{13}{10}, -\frac{13}{1}\right)$
 (c) $\frac{12}{13}\sqrt{26}$

11. Perpendicular to both planes.
 10. (a) $(5, -1, -2)$ (b) $\left(\frac{8}{19}, \frac{8}{13}, \frac{4}{13}\right)$

9. $\mathbf{r} = (2, 1, 0) + s(5, -8, -2) + t(3, 1, 1)$
 8. $\left(4, -\frac{7}{2}, \frac{2}{7}\right)$

- (h) $\mathbf{r} = (1, 1, 1) + t(5, 1, 1)$
 $z = 1 + t$
 $x = -2 + 3t$, $y = 3 - 7t + s$

- (g) $\mathbf{r} = (-2, 3, 1) + t(3, -7, 1) + s(0, 1, 0)$
 $y = 4 - t + 4s$, $z = 2 - t + s$

- (f) $\mathbf{r} = (1, 4, 2) + t(1, -1, -1) + s(2, 4, 1)$
 $x = 1 + t + 2s$
 $x = 2 + 3t + 6s$

- (d) $\mathbf{r} = (2, 3, -1) + t(3, -2, -3) + s(6, 5, -3)$
 $z = 1 + t - s$
 $x = 2 + 3t + 2s$, $y = -2t - 4s$

7. (c) $\mathbf{r} = (2, 0, 1) + t(3, -2, 1) + s(2, -4, -1)$
 $x = 2 + 3t + 2s$, $y = -2t - 4s$

- (h) $x + 2y - 7z + 4 = 0$
 (g) $x - 3z + 5 = 0$
 (f) $x - y + 2z - 1 = 0$

- (e) $x = 1$
 (d) $7x - 3y + 9z + 4 = 0$
 (c) $6x + 5y - 8z - 4 = 0$
 (b) $x + 3y - 3z + 5 = 0$
 6. (a) $x + 3y - 2z - 19 = 0$

21. For each of the following find the resultant and equilibrant of the given forces $\mathbf{F}_1$ and $\mathbf{F}_2$ where $\mathbf{F}_1$ and $\mathbf{F}_2$ act on the same particle.
- (a) $\mathbf{F}_1$ of magnitude 10 N acting due west
 $\mathbf{F}_2$ of magnitude 15 N acting due south
- (b) $\mathbf{F}_1$ of magnitude 8 N acting 240°
 $\mathbf{F}_2$ of magnitude 15 N acting 340°
22. A force $\mathbf{F}_1$ of 30 N acts on a particle together with a force $\mathbf{F}_2$ of 40 N. If the directions of $\mathbf{F}_1$ and $\mathbf{F}_2$ make an angle of 60° with each other, find their resultant and equilibrant.
23. Three forces of magnitude 30 N, 40 N, and 50 N act on a particle so that the particle is in equilibrium. Find the angle between the directions of the largest and smallest forces.
24. A sailboat is acted upon by the wind and water current. The wind velocity is 16 km/h east and the current velocity is 10 km/h north. Find the resultant of the two velocities.
25. A jet flies a course of 350° at 1000 km/h. If a 100 km/h wind is blowing from 90° determine the jet's ground speed and course.
15. Find the cosine of the angle between each of the following pairs of vectors.
- (a) $(1, 3), (2, 1)$ (d) $(6, 1, 2), (1, 2, -5)$
 (b) $(3, 4), (-1, -1)$ (e) $3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}, 2\mathbf{i} + \mathbf{j} + \mathbf{k}$
 (c) $(1, 3, 2), (1, -2, 4)$ (f) $2\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}, -2\mathbf{i} + 4\mathbf{j} + \mathbf{k}$
16. Which of the following pairs of vectors are perpendicular?
- (a) $(2, 3, 1)$ and $(1, 2, -8)$ (c) $(3, 4)$ and $(-\frac{1}{3}, \frac{1}{4})$
 (b) $(4, 1, 2)$ and $(1, -4, 2)$ (d) $(-2, -1)$ and $(1, -2)$

21. (a) 18 N, 214°
 18 N, 34°
 (b) 16 N, 310°
 16 N, 130°
22. 61 N, 35° from $\mathbf{F}_1$
 61 N, -145° from $\mathbf{F}_1$
23. 127°
24. 19 km/h, 58°
25. 1022 km/h, 344°

- (d) $-\frac{2}{\sqrt{1230}}$
 (e) $\frac{4}{\sqrt{174}}$
 (f) $-\frac{14}{\sqrt{357}}$
- (c) yes
 (d) yes

15. (a) $\frac{1}{\sqrt{2}}$
 (b) $-\frac{7}{5\sqrt{2}}$
 (c) $\frac{3}{7\sqrt{6}}$
16. (a) yes
 (b) no

16. Show that the line L of intersection of the planes

$$x - 3y + 7 = 0 \quad \text{and} \quad x - 2z - 4 = 0$$

has the following properties.

- (a) L is parallel to the plane $3x - 6y - 2z - 12 = 0$.
 (b) L lies in the plane $2x - 3y - 2z + 3 = 0$.
 (c) L is perpendicular to the plane $6x + 2y + 3z - 15 = 0$.

17. Find the cosine of the acute angle of intersection between each of the following pairs of planes.

(a) $2x + y - 3z + 7 = 0$ (b) $-x - y + 3z = 8$
 $4x - y + 7z + 5 = 0$ $2x - y + 4z = 1$

18. Find an equation for the plane that is parallel to both the line

$$\frac{x-1}{2} = \frac{y}{3} = \frac{z-1}{4} \quad \text{and the line} \quad \frac{x}{-1} = \frac{y+2}{2} = \frac{z+1}{1}$$

and passes through $(1, 0, -1)$.

19. Find an equation for the plane that contains the line

$$\frac{x+1}{4} = \frac{y}{-2} = \frac{z-2}{3} \quad \text{and is parallel to the line} \quad \frac{x}{3} = \frac{y}{4} = \frac{z+1}{-2}$$

20. Find a vector equation of the plane which contains the lines:

$$\frac{x-1}{4} = \frac{y+1}{2} = \frac{z-2}{3} \quad \text{and} \quad \frac{x-1}{5} = \frac{y+1}{4} = \frac{z-2}{3}$$

21. Prove that the four points $(1, 6, 3)$, $(-2, -4, -1)$, $(3, 9, 4)$, and $(-3, 0, 1)$ are coplanar.

22. A line through the origin, perpendicular to a plane, intersects the plane at the point $(1, 2, 3)$. Find an equation for the plane.

23. Given the planes whose equations are

$$(1 + k)x - 3y + 6z - 4 = 0$$

$$\text{and} \quad x + (5 + k)y + 3kz + 1 = 0$$

- (a) Find the value of k if the planes are parallel.
 (b) Find the value of k if the planes are perpendicular.
 (c) If the coefficient of z in the first equation was 5 instead of 6 could you solve parts (a) and (b)?

24. Show that the two lines

$$\frac{x+5}{3} = \frac{y-2}{2} = \frac{z+7}{6} \quad \text{and} \quad \frac{x}{1} = \frac{y+6}{-5} = \frac{z+3}{1}$$

lie in the same plane.

25. Find the equation of the line through the points $(1, 5, 1)$ and $(0, 4, 2)$. Show that the line lies on the plane

$$2x + y + 3z - 10 = 0.$$

15. (a) $\begin{pmatrix} 1 \\ 4 \\ -9 \end{pmatrix}$ (b) $\begin{pmatrix} -5 \\ -17 \\ 1 \end{pmatrix}$ (c) $\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$
 17. (a) $\frac{\sqrt{231}}{33}$ (b) $\frac{21}{21}$
 18. $r = (1, 0, -1) + s(2, 3, 4) + t(-1, 2, 1)$
 19. $r = (-1, 0, 2) + s(4, -2, 3) + t(3, 4, -2)$
 20. $r = (1, -1, 2) + s(4, -2, 3) + t(5, 4, 3)$
 21. Equation of plane is $2x - 5y + 11z - 5 = 0$.
 22. $x + 2y + 3z - 14 = 0$
 23. (a) -2 (b) $\frac{1}{2}$ (c) Part (a) could not be solved.
 24. Lines intersect at $(-2, 4, -1)$.
 25. $r = (1, 5, 1) + t(1, 1, -1)$: Coordinates of each point satisfy the equation of the plane.